

supervisor	
signature	

*You may answer in either Korean or English for problems 2~5. However, you should insert only English words in problem 1 and 6.

- Fred Brooks stated that over 90% of (a.) of a typical system arise in (b.) phase.
- (c.) means the ability to rely on the software to function properly when needed.
- Using global variables is dangerous mainly because this makes all modules (d.) one another.
- Constructor is a special member function that is automatically called when (e.) is (f.).
The name of a constructor is the same as the name of (g.).
Normally, you should put code in a constructor to (h.) member variables of a class.

benefits	explanation for each benefit with sufficient details
(i)	
(ii)	
(iii)	

(b) The amount of overheads is very small (i.e. almost negligible) when using call-by-reference mechanism for passing large objects into a function. Explain why? ()

```
1: #include <iostream>
2: using namespace std;
3: class Rectangle {
4:     int width, height;
5: public :
6:     Rectangle(int x,int y) : width(x),height(y){}
7:     int area() { return width*height; }
8: };
9: int main() {
10:     Rectangle a(3,4);
11:     Rectangle *b, *c;
12:     b = a;
13:     c = new Rectangle (5,6);
14:     cout << "area:" << a.area() << "\n";
15:     cout << "area:" << b.area() << "\n";
16:     cout << "area:" << c.area() << "\n";
17:     delete b;
18:     delete c;
19:     return 0;
20: }
```

(i) mark wrong part - 1 : (use arrow and circle for marking)
 why is it wrong? : ()
 modify it for correction. : ()

(ii) mark wrong part - 2 : (use arrow and circle for marking)
 why is it wrong? : ()
 modify it for correction. : ()

(iii) mark wrong part - 3 : (use arrow and circle for marking)
 why is it wrong? : ()
 modify it for correction. : ()

(iv) mark wrong part - 4 : (use arrow and circle for marking)
 why is it wrong? : ()
 modify it for correction. : ()

5. (33points) Consider an integer stack class **TIntStack** (posted on our class webpage) that is defined as follows.

```
-----
#include <iostream>

const unsigned int DEFAULT_SIZE = 256;

class TIntStack {
public :
    TIntStack(unsigned int stackSize = DEFAULT_SIZE);
    TIntStack(const TIntStack& that);
    TIntStack& operator=(const TIntStack& assign);
    ~TIntStack();
    void Push(int thisValue);
    int Pop();
    unsigned int HowMany() const;

private :
    int* _sp; // memory is dynamically allocated
            // during constructor function call
    unsigned int _count; // number of elements in stack
    unsigned int _size; // size (capacity) of stack
};
-----
```

(1) Write appropriate C++ code for following copy constructor.

TIntStack::TIntStack(const TIntStack& that)

```
{
}
}
```

(2) Write an appropriate C++ code for destructor.

TIntStack::~~TIntStack()

```
{
}
}
```

(3) Write an appropriate C++ code for member function **Push**

void TIntStack::Push(int thisValue)

```
{
}
}
```

(4) What problems occur if the copy constructor **TIntStack(const TIntStack& that)** is not provided (i.e. is not written) by class developers in the above source code of the class **TIntStack**? Write down your answers with sufficient details. (i.e. you have to first explain what happens and then explain why that is a problem.)

- what happens? ()
- why that is a problem? ()

6. (12 points) Fill out empty blanks with appropriate English words.

(a) Copy constructor is automatically called when ().

(b) There are mainly three different cases where a copy constructor of an object is called. Write an example for each case.

- case 1 example: ()
- case 2 example: ()
- case 3 example: ()